Multidimensional Stock Market Liquidity and Business Cycles*

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Abstract

Recognizing the difficulty in measuring stock market liquidity and its multidimensional nature, we comprehensively analyze the impact of multidimensional stock market liquidity on business cycles. Using the cross-sectional information contained in 7 different liquidity measures and applying a factor model approach, we find that stock market liquidity includes information on the future of the US. However, this information can differ significantly per liquidity dimension. We find evidence that the illiquidity ratio and the trading volume are better suited to capture information about the macroeconomy. In addition, we show that US liquidity measured by the illiquidity ratio and trading volume also contains information about the state of the U.K., Canadian, and Australian economy. Finally, we show that the commonality in liquidity cannot fully explain the relationship between stock market liquidity and the real economy. Our findings exemplify the importance of liquidity proxies in explaining the state of the macroeconomy.

Keywords: Stock Market Liquidity, Business Cycles, US Economy, Factor Models

Highlights:

- Liquidity measures differ in their information content about the real economy.
- Price-impact and trade volume measures can predict US GDP growth.
- Aggregate liquidity contains relevant information about the future economy.
- Aggregate liquidity is less predictive than some individual liquidity measures.
- Our results are consistent with the liquidity shock hypothesis.

1 Introduction

During the global financial crisis, the crucial role of market liquidity for effective market functioning and global financial stability was first emphasized. In the aftermath, several studies found that market liquidity and the real economy are connected. For example, Næs et al. (2011) shows that liquidity stock market liquidity can be used to forecast real economic GDP, whereas Ellington (2018) analyses the effect of liquidity shocks in the UK. Similarly, Galariotis and Giouvris (2015) analyzes the impact on national and global liquidity in different countries. They show that the relationship

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between liquidity and the real economy is not uniform and can be market-specific. In contrast, the results of Inekwe (2020) emphasizes the importance of market liquidity in the intertwinement of international economies. Their results indicate that liquidity spillovers can explain the comovement in real economic output between different countries.

These previous studies focus on a single or limited number of liquidity measures in their empirical analysis. However, market liquidity is a multidimensional concept and the relevance of some of the key characteristics of market liquidity can change over time, depending on periods of stability versus periods of stress. As noted in Amihud et al. (2006), a single stock market liquidity measure cannot capture all liquidity dimensions. Knowing if there is a difference between liquidity measures and dimensions in capturing information about the real economy could improve our understanding of the empirically established relationship. In addition, it might help us to understand the transmission of liquidity shocks across countries. Similarly, it might help investors and policymakers in choosing the appropriate liquidity measures.

In this paper, we further explore the relationship between the real economy and stock market liquidity, focusing on its multidimensional nature. Specifically, this paper uses the multifaceted information content of 7 liquidity proxies representing different liquidity dimensions to explain economic activity. We start by examining the impact of different liquidity measures on business cycles in the US economy from 1960 to 2022. By focusing on the U.S., we limit ourselves to an efficient and well-functioning equity market. Therefore, our results highlight the impact of stock market liquidity rather and we are able to (partially) avoid debating effects of the functioning of the chosen market. Secondly, we analyze the effect of US liquidity on the real economies of Canada and the United Kingdom.

Following Korajczyk and Sadka (2008) and Hallin et al. (2011), we use a factor model approach to summarize cross-sectional information for individual liquidity measures. Using a factor model, we can summarize the information from large panels of data and the comovements of the series in the panel. A factor model is an alternative to what is typically done in empirical studies of liquidity computing cross-sectional averages. The main advantage of using a factor model over cross-sectional averages is that the method grants us more freedom in deciding the number of one-dimensional indexes (the factors) we deem to represent the dataset appropriately. Secondly, by using a factor model approach, we can avoid the discussion that might arise in aggregating individual liquidity measures using cross-sectional averages. By expressing the original data set as the linear combination of a handful of unobserved factors to be appropriately estimated, we can aggregate cross-sectional units by directly exploiting the series' covariations. To explore the relationship between the different liquidity measures and the macroeconomic aggregates of interest, we broadly follow the approach of McCracken and Ng (2020) and that of Næs et al. (2011).

Our findings indicate that liquidity can contain relevant information about the future state of the US economy. However, this information can differ significantly depending on the chosen liquidity measure. We show that the illiquidity ratio by Amihud (2002) and trading volume seem most equipped to explain the future state of the economy. Similarly, we find that US liquidity measured by the illiquidity ratio and trading volume can be used to explain the future of the Canadian and U.K. economies, whereas other liquidity measures are generally not able to provide additional information. These results underline the finding of Næs et al. (2011) that the illiquidity ratio can be used to forecast economic growth. However, we extend their analysis by showing that only the illiquidity ratio and the trading volume can be reliable used to predict the state of the real economy. The fact that the illiquidity ratio contains relevant information is consistent with the liquidity shock hypothesis of Kiyotaki and Moore (2019). Furthermore, we find no robust evidence that direct trading costs contain significant information on future economic growth. Linking these channels to specific liquidity characteristics and/or measures is challenging. Our results indicate that liquidity

proxies that both directly capture aspects of the underlying price of the securities and the amount traded can be used to forecast growth. Although liquidity characteristics and measures can differ significantly, they still try to capture the same underlying concept. Consequently, one could argue that, in principle, focusing on different liquidity measures should not alter the observed relation between liquidity and the real economy. In the last section of our paper, we construct a measure of aggregate liquidity to inspect this issue. By creating an aggregate measure of liquidity, we can obtain a one-dimensional liquidity index that represents an underlying and unobserved liquidity process for the economy as a whole. We rely on creating an aggregate liquidity index as defined in Korajczyk and Sadka (2008). It allows us to summarize cross-sectional information for a single liquidity measure and, simultaneously, to have a statistical criterion to construct a measure of aggregate liquidity (by definition, incorporating information of all the considered liquidity measures). Using the aggregate measure, we analyze whether the commonality in liquidity measures drives the relationship between liquidity and business cycles. Our results show mixed evidence for a relationship between aggregate liquidity and real GDP. In-sample, we find strong evidence that aggregate liquidity contains relevant information about the future economy. However, aggregate liquidity cannot be used to forecast the real state of the economy more accurately than other liquidity measures.

In the next section (2), we discuss the theoretical relationship between stock market liquidity and the real economy in more detail and introduce the different measures of stock market liquidity and their dimensions. In Section 3, we introduce the description of the data, the factor model, and some descriptive statistics. Section 4 focuses on our empirical methodology, while in Section 5 we discuss the results of individual stock market liquidity proxies. Before concluding, in Section 6 we introduce the aggregate liquidity and the related results.

2 Real Economy & Stock Market Liquidity

There exist three main theories linking stock market liquidity to the real economy. First, there is the liquidity shock hypothesis introduced by Kiyotaki and Moore (2019). This hypothesis states that a sudden decrease in asset market liquidity leads to a corresponding reduction in stock price. Consequently, investment and output decrease as firms are hindered in financing due to a binding collateral constraint. Secondly, there is the investment channel. This channel states that higher transaction costs raise the cost of capital for firms, thereby hindering their investments. A more liquid market, on the other hand, stimulates investments in illiquid long-term projects. Levine (1991) describe this in more detail where costs on market transactions might hinder growth through the reallocation of resources by firms and households. Finally, stock market liquidity can reveal the information set of investors and act as a signaling mechanism (Florackis et al., 2014). When investors hold negative future economic expectations, they re-balance their portfolio from risky assets, such as stocks, towards, safer assets, such as bond. This phenomenon is referred to as the flight-to-quality and closely related with a flight-to-liquidity (Longstaff, 2002). This latest phenomenon refers to a situation where some market participants suddenly prefer to hold highly liquid securities rather than less-liquid securities. The flight-to-liquidity situation occurs in times of increasing uncertainty and during these times investors become more risk-averse shifting their portfolios to safer assets, a flight-to-quality situation. Therefore, the flight-to-quality reflects the investors' negative expectations about the future, which can be observed through decreasing stock market liquidity. These theories generally focus on liquidity as a general latent variable and do not focus on their different characteristics of liquidity. Kyle (1985) identifies three liquidity characteristics: tightness, resilience, and depth, while Sarr and Lybek (2002) also add immediacy and breadth as key characteristics. Tightness refers to the cost of trading, depth to the ability of the market to absorb quantities with-

out having a large effect on the price, resiliency to the speed with which prices tend to converge towards the underlying liquidation value of the commodity, immediacy to the speed with which orders are settled and executed, and breadth to the impact of large and high volume orders on the price. Liu (2006) link these characteristics to four liquidity dimensions: trading quantity, trading speed, trading cost, and price impact. Trading quantity refers to how many securities can be traded at a given price, trading speed to how quickly a security can be traded, trading cost to the actual costs of a given trade, and price impact to the ease with which a security can be traded with a minimum impact on price. Price-impact measures are often linked to the liquidity shock hypotheses. The hypothesis focuses on the resaleability of assets, whereas price impact measures focus on the impact of order flow changes on asset prices. If stock prices are more sensitive to the amount traded, it hinders the resaleability of the assets. Similarly, Levine (1991) links direct trading costs to the investment channel. As trading costs directly focus on capturing the explicit transactions costs of trading. Finally, in a model were a set of investors have a wider information set Vayanos and Wang (2013) show that price-impact is higher and that there is an ambiguous price-reversal effect. A downside of this model is that frictions that hinder trading and results in illiquidity are primitives rather than equilibrium outcomes. Another branch of literature that models market liquidity as an equilibrium outcome stems from monetary theory (see for example, Nosal and Rocheteau (2011) for a survey). For example, Lagos and Zhang (2020) use a monetary model with search and bargaining to link trading volume and bid-ask prices to micro-structure of the asset market (bargaining power and matching efficiency). An increase in market power of dealers results to an increase in trading volume and in an increase (decrease) in the bid (ask) price. However, they do not directly focus on the direct link between the real economy and stock market illiquidity. To summarize, there is plenty evidence that the price-impact measures and transaction costs measures capture relevant information about stock market liquidity. However, all theories focus on liquidity as a general latent variable. Therefore, we argue that in principle different liquidity measures should not differ in their ability to capture information about the real economy. Combined with our discussion above, this allows us to state the following loosely *liquidity measures should not differ in their ability to capture information about the real economy.*

Liquidity Measures

In this paper, we limit ourselves to choosing low-frequency liquidity measures that are constructed using widely available data, that are robust for different trading periods, and those that can be constructed quarterly for a long time series (Gabrielsen et al., 2012); (Garabedian and Inghelbrecht, 2015). In the literature, a wide set of different liquidity measures is used. Le and Gregoriou (2020), Díaz and Escribano (2020), and Goyenko et al. (2009) provide surveys on different liquidity measures, their performance, and their dimension. In our model, we use measures of direct trading costs, of indirect trading, of trading volume, and of price impact. To capture direct transaction costs, we use the relative and high-low spread. The relative spread captures the transaction cost of trading a security measured by the bid-ask spread, whereas the high-low spread implicitly estimates the bid-ask spread. Marshall et al. (2012) show that the relative spread is considered the best benchmark to capture transaction costs, whereas Corwin and Schultz (2012) show that the high low spread is better able to capture the spread compared to other implicit spreads. To capture the implicit cost of trading, we use the zero measures introduced by Lesmond et al. (1999) and the FTH measure by Fong et al. (2017). The zeros measure utilizes zero-return trading days to proxy the marginal transaction costs for an investor, whereas the FTH provides a measure of the overall costs associated with trading. We use both the trading volume and the turnover ratio as a proxy for the amount

traded. The trading volume provides the dollar amount of shares over a specific period, whereas the turnover ratio captures the number of times outstanding shares change hands. Finally, the illiquidity ratio by Amihud (2002) is used as a price impact measure. The illiquidity ratio reflects the daily price response associated with one dollar of traded volume. The included liquidity measures are aimed to represent a diverse set of liquidity measures. In addition, we try to include liquidity measures used in the literature. For example, the Illiquidity ratio is used by Florackis et al. (2014) and Næs et al. (2011), while Næs et al. (2011) also includes the relative spread. Unfortunately, the chosen liquidity proxies will never represent a complete picture of liquidity and will contain a certain degree of arbitrary. All our liquidity measures capture illiquidity except for the turnover ratio and the Trading Volume. Table 1 provides an overview of the liquidity measures and their relevant dimensions, while in Appendix 7 we show the construction of the different liquidity measures. We construct the measures on a quarterly basis. Individual daily liquidity measures are converted to quarterly by taking the firm-specific average over the entire quarter.

Table 1, around here.

3 Data, Factor Model, & Descriptive Statistics

We introduced our different liquidity measures and their respective dimensions in the previous section. We used the quarter-time series average per firm to convert them to the quarterly observations. Our dataset comprises of quarterly data since we are mainly interested in the relationship between macroeconomic variables and liquidity in this paper. Daily prices, market returns, shares outstanding, and share volume traded are downloaded from the CRSP database.[1,2] To have a homogeneous sample, we restrict our analysis to common shares trading on the New York Stock Exchange (NYSE). Our data set is constructed such that our analysis can be performed for the period 1962-2022, implying that the underlying data might span a wider period.[3] Finally, we exclude penny stocks and stocks with a price above $1000. Furthermore, per liquidity measure, we only consider firms for which at least 45 daily observations per quarter are available, and, for all liquidity measures, at least 8 quarterly observations are available. Secondly, we winsorize the final quarterly liquidity measure at the 1st and 99th percentile to remove the effect of any outliers. Our macroeconomic data is real GDP for the U.S., Canada, and the U.K. obtained from the Federal Reserve Economic Database (FRED) of the Federal Reserve Bank of St. Louis. Our dataset consists of quarterly GDP observations and 4959 liquidity time series for each of the 7 liquidity measures we consider.

Consequently, we consider a large cross-section for each liquidity measure in the order of thousands of observations. Adopting a factor model perspective allows us to reduce this large data availability to a smaller number of latent factors, which can be considered indices summarizing all the relevant information about the co-movements of a large set of regressors. Following Korajczyk and Sadka (2008) original notation, $\boldsymbol{L}_t^i$ be the $n \times 1$ vector of observations on the i-th liquidity measure ($i = 1, 2, .., 8$) at time T, with n being our cross-sectional dimension. Assuming that the

[1]The variable share volume from the CRSP database is defined as the units of shares sold on a day. The variable is rounded to the nearest hundreds. To transform share volume to units of dollars sold on a day, the original daily share volume is multiplied by the corresponding price.

[2]In case the highest (lowest) ask (bid) price is not available, we use the closing ask (bid) price.

[3]Caution is required by including the COVID period. During this period, the authorities' extensive liquidity provision might have altered the relationship between the real economy and liquidity. Although interesting, the impact of this type of intervention is outside the scope of this paper. Moreover, as argued by Ng (2021), the estimation of factors could be affected. However, the results for the period 1962-2019 do not significantly differ. Therefore, we only show the results for the entire sample period.

data-generating process for $\boldsymbol{L}_t^i$ is an approximate factor model, we have that

$$\boldsymbol{L}_t^i = \boldsymbol{B}^i \boldsymbol{F}_t^i + \boldsymbol{\eta}_t^i, \tag{1}$$

where $\boldsymbol{F}_t^i$ represents the $r \times 1$ vector of the latent common factors, $\boldsymbol{B}^i$ is the $n \times r$ matrix of factor loadings, and $\boldsymbol{\epsilon}^i$ is an $n \times 1$ vector of idiosyncratic shocks, specific to the individual cross-sectional units. As demonstrated in Connor and Korajczyk (1986) first, and later in Stock and Watson (2002a) under more general theoretical conditions, the latent factors $\boldsymbol{F}_t^i$ can be consistently estimated by principal component analysis. When estimating the factors, we need to determine the number of significant factors r. To this end, we adopt the IC_{p2} criterion developed in Bai and Ng (2002). Our empirical study takes inspiration from the literature on factor models employed in predicting macroeconomic aggregates. For example, Stock and Watson (2002b) and Bai and Ng (2006) represent some essential references. This last paper shows the conditions under which the factors estimated by principal components can be used confidently as predictors in the context of regression analysis, neglecting the degree of uncertainty that the estimation of the factors necessarily involves.

As widely practiced in the literature, macroeconomic time series are appropriately transformed to achieve stationarity. We transform the individual liquidity series by taking either logarithmic first differences or first differences depending on the nature of the liquidity measure. The choice is not irrelevant, given that in the literature, there are examples of works that alternatively use liquidity in levels and growth rates (consider Hasbrouck and Seppi (2001) and Chordia et al. (2000), respectively). We opted for the latter option primarily to deal with the fact that liquidity measures present signs of non-stationarity. Although it is theoretically possible to use factors calculated from non-stationary variables, we followed the most common practice in the literature and considered only the stationary case.

As is well known, estimated factors are characterized by sign indeterminacy. We decided that each factor represents liquidity rather than illiquidity. Therefore, similar to Korajczyk and Sadka (2008), the sign of the factor estimated for each measure of illiquidity (all except the turnover ratio and trading volume) is chosen so that it is negatively correlated with the time series of the cross-sectional mean of that measure. In contrast, the signs of the factors estimated for the measures of liquidity are chosen so that the correlation is positive. Indicating that an increase in the factor is associated with a positive growth in liquidity.

After transforming the series, the available data range from the first quarter of 1960 to the last quarter of 2018, for 236 time-series observations. To address the significant presence of missing values in the data set that collects liquidity measures of interest, we adopted the EM algorithm described in Stock and Watson (2002b) in the re-worked form proposed by McCracken and Ng (2020). In this case, the missing values are imputed by the algorithm in the procedure of estimating the factor model. To ensure the correct functioning of the procedure, we considered it correct to control the size of the data panel used, limiting ourselves to only including the historical series of individual companies for which the total number of missing values does not exceed a particular percentage share of the size of the considered estimation window itself. We adopted this approach because observations for most companies in our sample are available only for a subset of our time horizon. Therefore, they contain only substantial information content during this period. Moreover, this allows us to (partly) mitigate survivorship bias; by not simply eliminating firms that are not present during the entire sample period, we still exhaust the information for firms that are not present during the entire sample. Table 2 shows the trade-off between the number of included stocks and the number of missing observations for different quotas using the entire sample as an estimation window. The table indicates a trade-off between the number of missing observations that need to be

imputed and the cross-sectional dimension. Setting the quota requires cautiousness to ensure that the cross-sectional dimension is sufficiently large. For factors estimated using the entire sample, we use a quota of 60%. Our empirical evidence suggests that altering the quota does not alter the results in a significant manner; it is this reduced group of cross-sectional units that contains the relevant information to estimate the factors.

Table 2 around here.

Next, we present some descriptive statistics related to the factors extracted for each of the seven individual liquidity measures using the entire available sample. As a first step, the factors for each liquidity measure are estimated via principal component analysis using the EM algorithm to account for numerous missing values. The appropriate number of relevant factors for each measure is estimated using the IC_{p2} criterion discussed Bai and Ng (2002). We allowed a maximum of five factors in the estimation process with this criterion. Figure 2 shows the time series of the estimated factors.

Figure 2, around here.

In particular, two factors are selected for the illiquidity ratio and trading volume series. On the other hand, three distinct factors are selected for the spread series, meaning that a larger dimensionality characterizes the latent factor space. In general, all the series exhibit a degree of time-varying volatility. This fact is especially evident in the case of Zeros and FHT factors, as their series shows a significant reduction in their volatility in the early 2000s. It is well known that when the factors are estimated via principal component analysis, they have a natural ordering corresponding to the fraction of the total variance of the original series that they can explain. We report the share of the explained total variance in Figure 3 to provide a perspective on the informativeness of the first few estimated factors. In most cases, the first factor is predominant above all others, explaining a large portion of the total variance of the original series, and subsequent factors appear to be relatively less informative in this sense. Nevertheless, the information content of the remaining factors is non-negligible as the bars displayed in the plots approach zero slowly enough. This exemplifies why the IC_{p2} criterion selects more than one factor (see Figure 2), even in cases where the fraction of variance explained by the first factor is prominent compared to the others. In the case of the ZEROS and FHT series, for both of which we select a single factor, the estimated factors explain just a marginal portion of the variance of the underlying series. We can also note how reduced information content characterizes the factors estimated for the FHT and ZEROS series, indicating that a factor model may not be the best way to capture the characteristics of these data sets. This fact could be imputable to the abrupt change in behavior in these series highlighted in Figure 2, resulting in a more complex behavior that the factor representation can capture with greater difficulty compared to what we observe for other measures. To rule out the presence of non-stationary patterns in the factor series, the autocorrelation functions computed for the first estimated factor for each measure are also reported in Figure 4. We can see how all factors exhibit only moderate signs of autocorrelation.

Figure 3, around here.

Figure 4, around here.

Finally, we report lead-lag canonical correlations between the group of factors estimated for the series of each liquidity dimension to examine the patterns characterizing the relationships between the different groups of factors at various lags. Figure 1 reports the first canonical correlation (contemporaneous) between each group of common factors at multiple lags, providing us with a multivariate dependence measure for the estimated factors' series. We can notice how some liquidity measures exhibit signs of a stronger correlation with some specific measures rather than with other liquidity dimensions. For instance, the trading volume factors show strong contemporaneous corre-

lations with both the illiquidity ratio and the turnover ratio factors. Similarly, the factors computed for the different spread measures series appear strongly correlated at all considered lags. Despite this, we are led to exclude the possibility that a single liquidity measure drives the others, as they all exhibit a certain degree of independence in their behavior. This is especially important for our analysis because, in the case of lead-lag relationships between the different liquidity measures, we would not be able to pinpoint an accurate effect.[4]

Figure 1, around here.

4 Empirical Model

For our empirical model, we follow the framework adopted by Næs et al. (2011), focusing on whether liquidity contains relevant information about the future real GDP growth and splitting our empirical study between an in-sample and an out-of-sample analysis. Let y_t denote the first logarithm difference of the variable of interest and denote the same variable to be predicted in a h-period ahead forecast as y_{t+h}^h. In our case y_{t+h}^h is defined as the variable's h-period growth at a quarterly rate

$$y_{t+h}^h = \frac{\ln\left(y_{t+h}/y_t\right)}{h}. \tag{2}$$

where h is the forecast horizon considered. Throughout the paper, we consider predictions over three different forecast horizons: one-quarter ahead, two-quarters ahead, and four-quarters ($h = 1, 2, 4$). In our analysis, we consider a regression equation of the form:

$$y_{t+h}^h = \alpha_h + \boldsymbol{\beta}_h^i\left(L\right)\boldsymbol{F}_t^i + \boldsymbol{\gamma}_h\left(L\right)y_t + \epsilon_{t+h}^h, \tag{3}$$

where i denotes the individual liquidity measure, $\boldsymbol{F}_t^i$ the r dimensional factor, with $\boldsymbol{\beta}_h^i\left(L\right)$ and $\boldsymbol{\gamma}_h\left(L\right)$ being finite order lag polynomials. In particular, $\boldsymbol{\beta}_h^i\left(L\right)$ is our coefficient of interest, summarizing the information captured by liquidity about future economic conditions. In our in-sample analysis, we are interested in the regression coefficient that links our liquidity factors with the macroeconomic variable. To compare the information content across the different liquidity measures, we use the adjusted R-squared, AIC, and BIC. For example, as explained by Spiegelhalter et al. (2014), these information criteria capture different ideas: the AIC identifies a model that can make good short-term predictions, whereas the BIC is trying to identify the 'true' model. Secondly, we conducted a Granger causal analysis. Before performing the Granger Causality tests, the optimal number of lags for the underlying vector autoregressive model is chosen using Akaike's Information Criteria (AIC). We apply a Block Granger Causality test to check if excluding all liquidity factors in predicting real GDP provides evidence of causality. Secondly, we test for reverse causality for each time series factor by excluding GDP from their regression equation. Next, we perform a pseudo-out-of-sample analysis. Generally, a pseudo-out-of-sample analysis simulates a forecaster's experience using a prediction model when the available sample is progressively updated by including the new data over time in the series. The forecasting procedure proceeds as follows. At each point in time (for each quarter, in our case), we consider a certain window of data of finite dimension ω so that the sample used in the estimation process ranges from time $t-\omega+1$ to the current time t. We use the available information at time t to predict the h-th step forward value of the variable y_t. This preliminary step requires obtaining estimates of the process $\widehat{\boldsymbol{F}}_t^i$ to be used in the forecasting model so that we obtain a different factor estimate for each window. We limit ourselves to using a maximum estimated

[4]It is important to note that we are considering lead-lag relationships on a quarterly horizon.

number of three factors chosen by the above-mentioned information criteria in the out-of-sample analysis. We follow the maximum number of factors selected in-sample to simplify any discussion about the forecast model adopted and avoid any indecision regarding the number of factors necessary to provide a correct approximation of the factor space. Based on the estimated factors, we adopt the following general forecasting equation.

$$\widehat{y}_{t+h}^{h,m} = \widehat{\alpha}_h + \sum_{l=1}^{\widehat{r}} \sum_{k=1}^{2} \widehat{\beta}_{hk,l}^{i} \widehat{F}_{l,t-k+1}^{i} + \sum_{j=2}^{4} \widehat{\gamma}_{hj} y_{t-j+1}, \tag{4}$$

where $\hat{r}$ refers to the estimated number of factors. The choice over the number of lags to be considered in the autoregressive structure of the GDP growth follows McCracken and Ng (2020) as we include our dependent variable up to its fourth lag. Concerning the estimated factor process, we include it up to its first lag. Notice that we do not include the first lag of y_t in the equation. This is motivated by the fact that these data are not readily available for every quarter and are provided by the relevant authorities with a delay of a quarter each time. Adopting the point of view of a market operator who wants to make forecasts about a specific macroeconomic aggregate, the information available to him stops in effect at the previous quarter compared to the one in which the forecasts are made. This is not the case concerning the different liquidity measures. These are available in real-time and ready for factor estimation and prediction. This point regarding estimating factors should be stressed, given that factors are unobservable variables that must be calculated following an appropriate methodology. In this sense, our methodology for GDP prediction is fundamentally based on two steps, given that as a first step, we obtain an estimate of the factors and subsequently include the factors thus estimated in the regression model. As discussed before, the presence of missing values is accounted for by the use of the EM algorithm described in McCracken and Ng (2020) augmented with a quota describing the particular percentage share of the size of the window that the total number of missing values does not exceed. For this reason, the number of individual cross-sectional units used in the single data window for factor estimation is not constant over time as different windows are considered. The results reported in this section for the out-of-sample analysis are obtained for a quota requirement for each company of at least 80% of the observation in the window being not missing values. This greatly reduces the number of companies that are actually represented in the cross-section for each window. We report the mean, median, maximum, and minimum of the cross-sectional observations used in a single window in table 3. As we can notice from this table, the number of cross-sectional units considered in the different windows does not undergo significant variations over time, and the oscillations around its average appear limited. This fact, therefore, seems to imply that a small number of companies are relatively well represented in the sample, so on average, the lowest number of missing values is observed. Our empirical evidence suggests that altering the quota does not alter the results in a significant manner; it is this reduced group of cross-sectional units that contains the relevant information to estimate the factors.

Table 3, around here.

Our forecast analysis compares the predictive accuracy of the different liquidity measures. All of our estimates are obtained employing a rolling window of 120 quarters. As a first step, we obtain a sub-dimension of the initial cross-section by removing all units for which the missing values exceed a pre-specified quota of the total size of the window. Secondly, using the remaining panel, we estimate the factors via principal component analysis. Third, the estimated factors are used in Equation 4 to generate forecasts of the variable y_t^h. The predictive performance of the models is finally evaluated

in terms of mean square forecast error, $MSFE_h$, defined as

$$MSFE_h = (T_1 - T_0)^{-1} \sum_{\tau=T_0+h}^{T_1} \left(y_\tau - \hat{y}_\tau^h\right)^2, \tag{5}$$

where T_0 is the first point in time for out-of-sample evaluation and T_1 is the last point in time for which we compute MSFE at horizon h. As before, we check whether our liquidity measures (estimated factors) contain relevant information about our macroeconomic variables by comparing their predictive accuracy. We compare an auto-regressive model with liquidity to a benchmark model represented by an auto-regressive model without including liquidity measures among the covariates to determine whether liquidity provides additional and unique information about the real economy. We define the benchmark model as the restricted model, whereas the benchmark model plus liquidity is the unrestricted model. Two test statistics are considered to test whether adding the liquidity measure to the benchmark model increases the accuracy of the forecast. The first test, the MSE-F test, tests for equal mean squared errors for nested models, while the second test, the ENC-NEW test, tests whether including stock market liquidity results in a higher predictive accuracy. These tests are introduced in Clark and McCracken (2001), who, together with McCracken (2007), also provide the bootstrap critical values. The first test is constructed as follows:

$$MSE - F = (P - h + 1)\frac{MSFE_r - MSFE_u}{MSFE_u} \tag{6}$$

where the $MSFE_r$ is mean squared forecast error of the restricted model, $MSFE_u$ of the unrestricted model, P the number of forecasted quarters, and h the forecast horizon. The second test statistic is calculated as follows:

$$ENC - NEW = (P - h + 1)\frac{P^{-1}\sum_t \left(\epsilon_{r,t+1}^2 - \epsilon_{r,t+1}\epsilon_{u,t+1}\right)}{MSFE_u} \tag{7}$$

where P is the number of quarters forecasted, h the forecast horizon,$\epsilon_{r,t+1}^2$ the squared error of the restricted model, $\epsilon_{r,t+1}$ and $\epsilon_{u,t+1}$ the error of the restricted and unrestricted model respectively, and $MSFE_u$ the mean squared forecast error of the unrestricted model.

The second part of the forecast analysis compares the prediction power of different stock market liquidity measures. We run the forecast equation and calculate the mean squared forecasting error. To compare the forecasting power between two stock market liquidity measures, the relative MSFEs are used, which is the MSFE of the first stock market liquidity measure, referred to as model 1, divided by the MSFE of the second stock market liquidity measure, model 2. The modified Diebold-Mariano test (Harvey et al., 1998) is used to test whether the difference of the MSEs between two predictors of the same dependent variable is different from zero. The modified Diebold-Mariano test is preferred over the Diebold-Mariano test as it is better suited for small samples. The test's null hypothesis is that two variables predicting the same dependent variable have an equal predictive accuracy; in other words, the MSFEs are equal. The two regressions with independent variables used to forecast are called model 1 and model 2, respectively. The test statistic is calculated as

follows:

$$d = P^{-1} \sum_{t} \left(\epsilon_{i,t+1}^2 - \epsilon_{j,t+1}^2 \right) \tag{8}$$

$$MDM = \sqrt{\frac{P + 1 - 2h + P^{-1}h(h-1)}{P}} \frac{d}{\sqrt{\frac{\gamma_d^2}{P}}} \tag{9}$$

where P is the number of out-of-sample quarterly forecasts, $\epsilon_{i,t+1}^2$ and $\epsilon_{j,t+1}^2$ the squared forecast errors of the two compared models i and j, h is the forecast horizon, and γ_d is $\gamma_0 + 2\sum_{k=1}^{h-1} \gamma_k$, where γ_{h-1} is the auto-covariance of d of order h-1. The MDM statistic follows a student's t-distribution with $P - 1$ degrees of freedom.

5 Results

We start by looking at the in-sample and out-of-sample results for the U.S., before showing the out-of-sample results for the U.K. and Canada.

5.1 In-Sample

Table 4 presents our regression results in sample, evidencing how the dollar volume and the illiquidity ratio measures provide information about the future real economy regardless of the forecast horizon considered. Furthermore, the high-low measure shows a significant coefficient when predicting GDP growth one or two quarters ahead. Inspecting the sign of the coefficients, we observe that a change in liquidity positively affects future economic output growth. This aligns with most theoretical predictions, where increased liquidity increases GDP growth. Finally, we compare the adjusted R-squared, the AIC, and the BIC to compare the fit of the different models. As our goal is not to create the best-suited forecasting model for GDP, we are agnostic about the actual fit. Still, we are instead interested in the difference between models that use different liquidity proxies. Again, the traded volume and the illiquidity ratio exhibit the best empirical fit for all forecasting horizons, with the illiquidity ratio showing slightly better performance, suggesting, therefore, that specific liquidity measures are better apt to explain the state of the US economy.

Table 4, around here.

To further explore whether liquidity can explain changes in real GDP, we provide the results of the Granger causality test in Table 5. The first row shows that Trading Volume and the Illiquidity Ratio Granger-cause economic growth. This is consistent with the results discussed above of our in-sample prediction. Furthermore, we find that the high-low spread Granger causes GDP growth, indicating that transaction costs could influence GDP growth. To explore the possibility of reverse causality in the relationship between GDP growth and changes in liquidity, we check for this eventuality in the last three rows of Table 5, showing no evidence that GDP causes changes in liquidity.

Table 5, around here.

5.2 Out-of-Sample

Our in-sample results indicate that stock market liquidity contains relevant information about the future state of the US real economy, with different liquidity measures that capture relevant information about future economic conditions. However, good in-sample predictions do not always guarantee good out-of-sample predictions. Table 6 contains the results for the out-of-sample forecasts. In table

6, we compare the predictive accuracy of the liquidity model versus an auto-regressive benchmark model. An MSE-ratio, the mean squared forecasting error of an AR-model plus liquidity divided by the mean squared forecasting error of the AR-model, below one, indicates that the model including liquidity results in a lower MSFE, whereas above one tells us that the model without liquidity performs better. The illiquidity ratio and the trading volume increase the predictive accuracy across the forecast horizon. Furthermore, the High-Low spread shows a ratio below one at a 1-quarter ahead forecast horizon,. Inspecting the MSE-F test, we find that these ratios are also significantly different. This provides more evidence that liquidity can contain information about the real economy. Finally, the ENC-NEW test provides evidence that these also encompass the AR model. Overall, our out-of-sample results again offer significant evidence that some liquidity measures contain relevant information about the future state of the U.S. economy. This information seems to depend on the choice of the liquidity measure. Similarly to the in-sample results, the trading volume and the illiquidity ratio seem especially relevant.

Table 6, around here.

Lastly, we want to highlight the importance of choosing a liquidity measure to explain real GDP. Table 7 compares the predictive accuracy of our different liquidity measures. The ratio shows the MSFE of model 1, column, divided by the MSE of model 2, row. For example, the first entry column relative spread and row traded volume show the MSFE of the relative spread divided by that of traded volume. This ratio is above one, indicating that trading volume results in a lower MSFE. The star shows that according to the modified Diebold-Mariano test, this difference is significant. We find that the illiquidity ratio outperforms all proxies when forecasting a quarter ahead, whereas the illiquidity ratio and trading volume outperform other liquidity measures for longer forecast horizons. This indicates that the illiquidity ratio and trading volume are the most suitable to explain GDP growth.

Table 7, around here.

5.3 Canada & United Kingdom

To highlight the importance of stock market liquidity in the international economy, we briefly focus on the impact on the U.S. liquidity and real GDP in Canada and the U.K. It is important to note that our focus in not in using liquidity as a Granger-cause for real British or Canadian GDP growth. Rather, we focus on the information content captured by different liquidity measures. Tables 8 and 9 contains the predictive accuracy of the liquidity model versus an auto-regressive benchmark model of real GDP growth. As before, an MSE-ratio below one indicates that the model including liquidity has a higher predictive accuracy, whereas a ratio above one tells us that the model without liquidity performs better. Inspecting, the table shows again that the illiquidity ratio and the trading volume increase the predictive accuracy across the forecast horizon also for Canadian and British real GDP growth. This highlights that also using U.S. liquidity in an international setting the choice of the liquidity proxy is important.

In general, our results indicate that different liquidity measures contain significantly different information about the state of the future economy. Thereby directly opposing our hypothesis. We find that both illiquidity ratio and trading volume are best equipped to forecast the future US economy. Compared to other measures, the trading volume and the illiquidity ratio have in common that they both contain elements of the amount traded and the price of the underlying security. The illiquidity ratio can be direclty linked to liquidity shock hypothesis of Kiyotaki and Moore (2019). Similarly, as trading volume captures the monetary amount effectively traded, one could argue that it partially captures a similar effect as the illiquidity ratio over a quarterly horizon. Thereby hinting that the relationship between stock market liquidity and the real economy is best

explained by the liquidity shock hypothesis. Another potential explanation is the fact that the illiquidity ratio measures disagreement between investors. If there is more disagreement between investors, one might argue that there is more uncertainty about the future, resulting in a flight to liquidity. Thereby supporting the signalling channel. Measures that do not directly focus on capturing the transaction cost seem better equipped to explain GDP growth. Besides the fact that the investment channel might not be play a crucial role, another possible explanation is that, as seen in most theoretical models of liquidity, the dealers set the bid-ask spread based on the expected behavior of investors (O'hara, 1998). In contrast, economic theories focus mainly on the behavior of investors. For example, the flight-to-liquidity channel is based on the investor information set. This may not necessarily be reflected in the bid-ask spread or the transaction costs investors incur, but it requires more detailed information (e.g. order book information). Our results for the illiquidity ratio are overall consistent with Næs et al. (2011): the illiquidity ratio contains relevant information about the future state of the US economy. However, using different liquidity measures, namely the implicit spread of Roll (1984) and the transaction cost measure of Lesmond et al. (1999), Næs et al. (2011) also finds that direct trading costs contain significant information about the future state of the economy. This difference might arise from the use of different measures, the use of a different time horizon, or the fact that we incorporate the entire cross-sectional firm's data.[5] This was already pointed out by Galariotis and Giouvris (2015) using the Roll's implicit spread and illiquidity ratio. Overall, this underlines the importance of choosing a suitable liquidity measure to explain the real economy.

6 Overall Liquidity

Our results for individual liquidity measures provide evidence that stock market liquidity measures differ significantly in their information content about the real economy. This might raise the question of whether liquidity itself contains relevant information about the future economy or specific characteristics picked up by an individual measure. Focusing on different liquidity measures should not alter the observed relation between liquidity and the real economy. We inspect this question by constructing a measure of aggregate liquidity and comparing its predictive power to the predictive power of the individual measures. Therefore, we adopt the definition of commonality in liquidity to indicate a multidimensional aspect of measured liquidity shared with the different individual measures discussed in this paper. The literature has examined various approaches for modeling this factor, which is common to the multiple dimensions of liquidity. We follow Korajczyk and Sadka (2008), using a factor model approach to capture the common information shared by the series of heterogeneous liquidity measures (defined in their work as across-measure liquidity). In addition to relying on all the liquidity measures of individual companies, the Korajczyk and Sadka (2008) methodology does not require any explicit assumption regarding the nature of the underlying data and the relationships between the different liquidity measures. The data relating to the various liquidity measures collected for a cross-section of companies are merged into a single high-dimensional panel. The co-movements of the entire cross-section are then modelled by a minimal number of factors representing the component commonly shared by the different measures included in the panel. However, this approach can present significant critical issues if the characteristics of the starting panel are not controlled. As discussed in Boivin and Ng (2006), increasing the dimensionality of the data can negatively affect factor estimates in some instances. This effect is significant when the

[5] Unfortunately, the remaining measures of Næs et al. (2011) are unsuitable for our factor model specification. There are too few cross-sectional observations for the implicit spread and the trading cost measure to estimate our factors credibly.

series under consideration have heterogeneous dimensions and different levels of clustering. From the point of view of the different liquidity measures that we consider in our dataset, some of these may be significantly more correlated than others (consider, for example, that some of the definitions of liquidity, although different, are almost perfectly overlapping). This can lead to various forms of inefficiency in factor estimation. For example, given two groups of series relating to liquidity measures which are by their definition strictly correlated, we could see that the estimated factor (which, remember, is modeled starting from the co-movements between the various series in the datasets) relies more heavily on these two dimensions of the panel rather than on the others.

Table 10, around here.

Table 10 reports the variance decomposition of the aggregate factors. It appears that, for all our liquidity measures, the common factor explains most of the variability of the underlying data. This highlights the common component in our different liquidity measures and indicates that all proxies try to capture the same latent process. In Tables 11 and 12, we show the in-sample results obtained using the across measure in our regression model. Testing for Granger causality, the aggregate illiquidity Granger causes future GDP growth, indicating that aggregate liquidity can be used to predict GDP growth. This is further established by the significance of the coefficients shown in Table 11. Comparing the goodness of fit with the individual liquidity proxies (Table 4), we find that the aggregate measure outperforms all other liquidity measures except the trading volume and the illiquidity ratio. Although the common factor can explain at least 60 percent of the variance of each measure, this alone does not seem sufficient to explain the observed relationship completely.

Table 11, around here.

Table 12 around here.

Inspecting the relationship between aggregate illiquidity and GDP growth out-of-sample further illustrates this complexity. The aggregate factor increases the predictive accuracy compared to an AR model across all forecasting horizons (Table 13), indicating that aggregate liquidity contains relevant information about the future state of the economy. Comparing the predictive power of the aggregate factor with individual factors, we find that the aggregate factor outperforms all measures except the illiquidity ratio and trading volume (see Table 14). Again, this indicates that the aggregate illiquidity factor contains relevant information about the future US economy, even though this information is captured mainly by the illiquidity ratio and trading volume.

Table 13, around here.

Table 14, around here.

7 Conclusion

This paper examines the impact of multidimensional stock market liquidity on business cycles in the US economy from 1962 to 2022, which captures the key market liquidity characteristics. In particular, we examine whether stock market liquidity measures differ in their prediction of business cycles. In contrast to the literature, we use the information content of multidimensional liquidity proxies to fully understand the importance and relevance of market liquidity. We used seven different liquidity measures and relied on a factor model approach to find that stock market liquidity can act as an indicator of future economic conditions. However, individual proxies of liquidity differ significantly in their prediction accuracy. Our predictions indicate that the illiquidity ratio provides robust information about GDP growth. This provides some supporting evidence for the stock liquidity hypothesis of Kiyotaki and Moore (2019), highlighting the importance of the resaleability of assets. Finally, by constructing aggregate liquidity measures, we can obtain a one-dimensional liquidity index that represents an underlying and unobserved liquidity process for the economy as a

whole. Using these indexes, we find mixed evidence that the common component of liquidity proxies is the driving force behind the observed relationship between stock market liquidity and the real economy. Before we conclude, we first emphasize that the usage of factor models in the construction of liquidity indexes can be improved by examining the nature and extent of contributions made by specific sub-dimensions of the cross-section to the factor estimates. In a large panel context, it is valuable to identify which firms or subsamples contribute the most significantly to the estimated factors. This understanding can only be achieved by closely analyzing the characteristics of the matrices of the factor loadings. Although this topic goes beyond the scope of this paper, which primarily focuses on the relationship between liquidity and real economic dynamics, an analysis of the factors and their loadings related to specific liquidity dimensions could provide deeper insights into the multifaceted nature of liquidity. This would contribute to the extensive literature dedicated to this subject. Secondly, we would like to stress that even though our analysis is limited to the U.S., our main result, that liquidity measures differ in their information content on the real economy, is a more general message. For example, as stock market liquidity contain information about the real economy for less developed financial market (e.g. (Si et al., 2019) for China), performing a similar analysis might provide new insights in the relation between the functioning of financial markets and the real economy. In general, we corroborate the conclusion of Næs et al. (2011) that stock market liquidity can contain helpful information to estimate the future state of the economy but mainly exemplifies the importance of liquidity dimensions and the choice of liquidity measure to explain the state of the economy.

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Table 1: Liquidity Measures and Dimensions

Liquidity Proxy	Background	Dimension
Relative Spread	Transaction Costs/Bid-Ask Spread	Tightness
Trading Volume	Trading Activity	Price Impact, Bread, Depth and Volume
Turnover Ratio	Frequency of Trading	Bread, Depth and Volume
High-Low	Transaction Costs/Implicit	Tightness
Illiquidity Ratio	Price Impact of Trading	Price Impact and Volume
Zeros	Marginal Transaction Costs	Tightness
FTH	Trading Cost	Tightness

This table contains a brief description of the background and the main dimensions of the liquidity measures employed throughout the paper.

Table 2: Example of Quota

Quota	90%	80%	70%	60%	50%	40%	30%	20%	10%	0 %
NA %	2.07%	6.48%	11.08%	19.81%	26.88%	38.08%	47.33%	55.85%	65.59%	74.08%
Total Firms	158	234	314	488	677	1107	1686	2430	3615	5169

This table illustrates the use of the described quota to deal with missing observations. The table shows the share of missing observations in our sample for the relative spread and the amount of cross-sectional observations (total firms) for different percentiles of the quota.

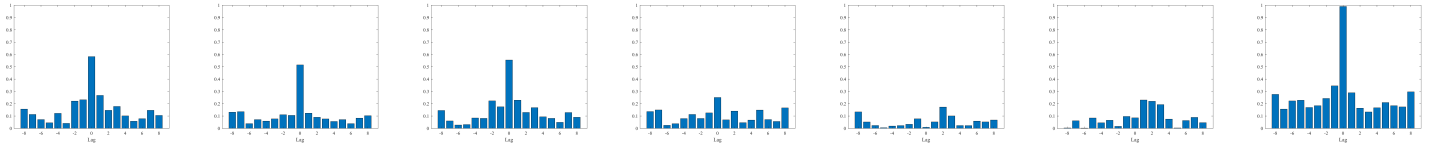
Table 3: Out-of-Sample: number of cross-sectional observations

	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
mean	528	528	528	528	517	538	528
median	524	524	524	524	519	541	524
max	612	611	611	612	598	621	612
min	435	435	435	435	433	445	435

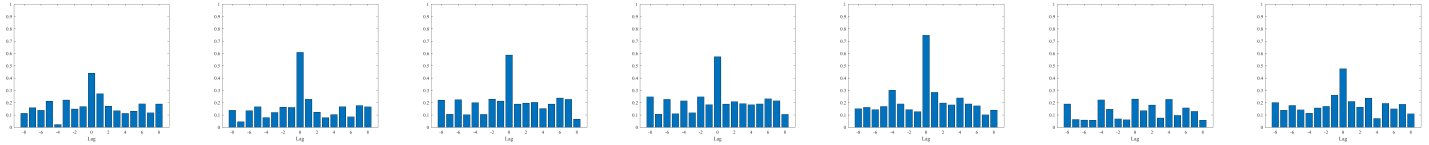
This table shows the cross-sectional observations used in our rolling window approach. The first row contains the rounded mean of cross-sectional observations, the second the median, and the last two rows respectively the minimum and maximum number of cross-sectional observations used in a specific time window. We report these statistics for our 7 different liquidity measures.

Figure 1: Lead-Lag canonical correlations

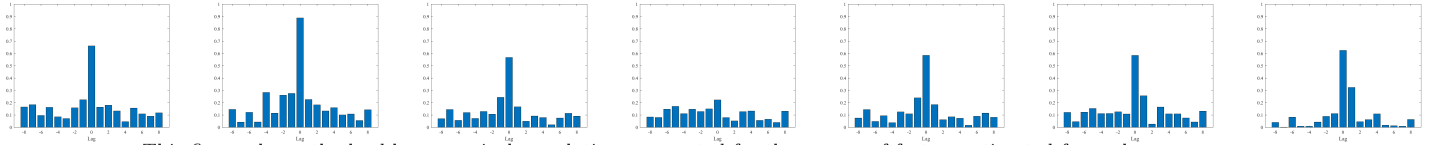
(a) FHT- HLS (b) FHT-ILR (c) FHT-RS (d) FHT-TV (e) FHT-TR (f) FHT-ZEROS (g) HLS-RS



(h) HLS-TV (i) HLS-TR (j) ILR-HLS (k) ILR-RS (l) ILR-TV (m) ILR-TR (n) TV-RS

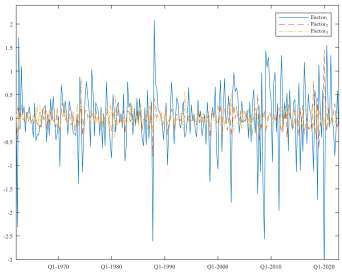


(o) TR-RS (p) TR-TV (q) ZEROS-HLS (r) ZEROS-ILR (s) ZEROS-RS (t) ZEROS-TV (u) ZEROS-TR

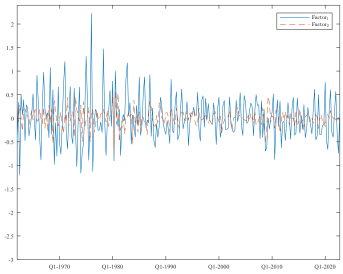


This figure shows the lead-lag canonical correlations computed for the groups of factors estimated for each measure.

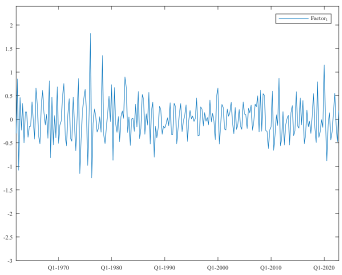
(a) Relative Spread



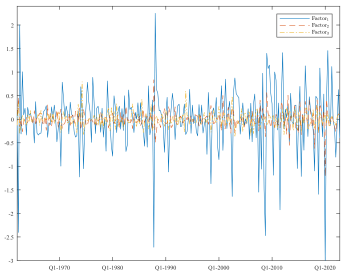
(b) Trading Volume



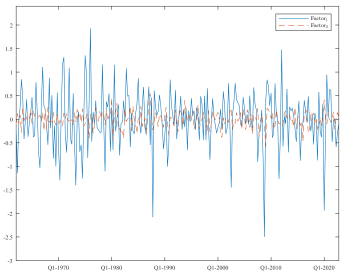
(c) Turnover Ratio



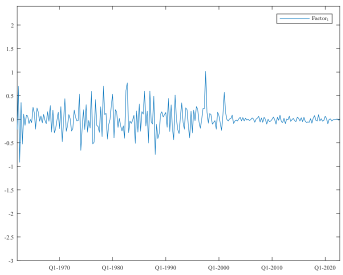
(d) High-LoW Spread



(e) Illiquidity Ratio



(f) Zeros



(g) FHT

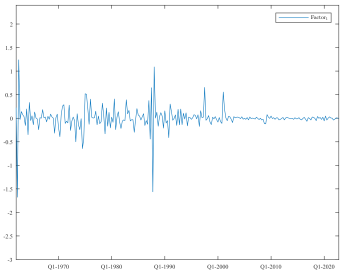


Figure 2: Times series plot of estimated factors for the interval Q1 1962 to Q4 2021. For each liquidity dimension we select up to three factors.

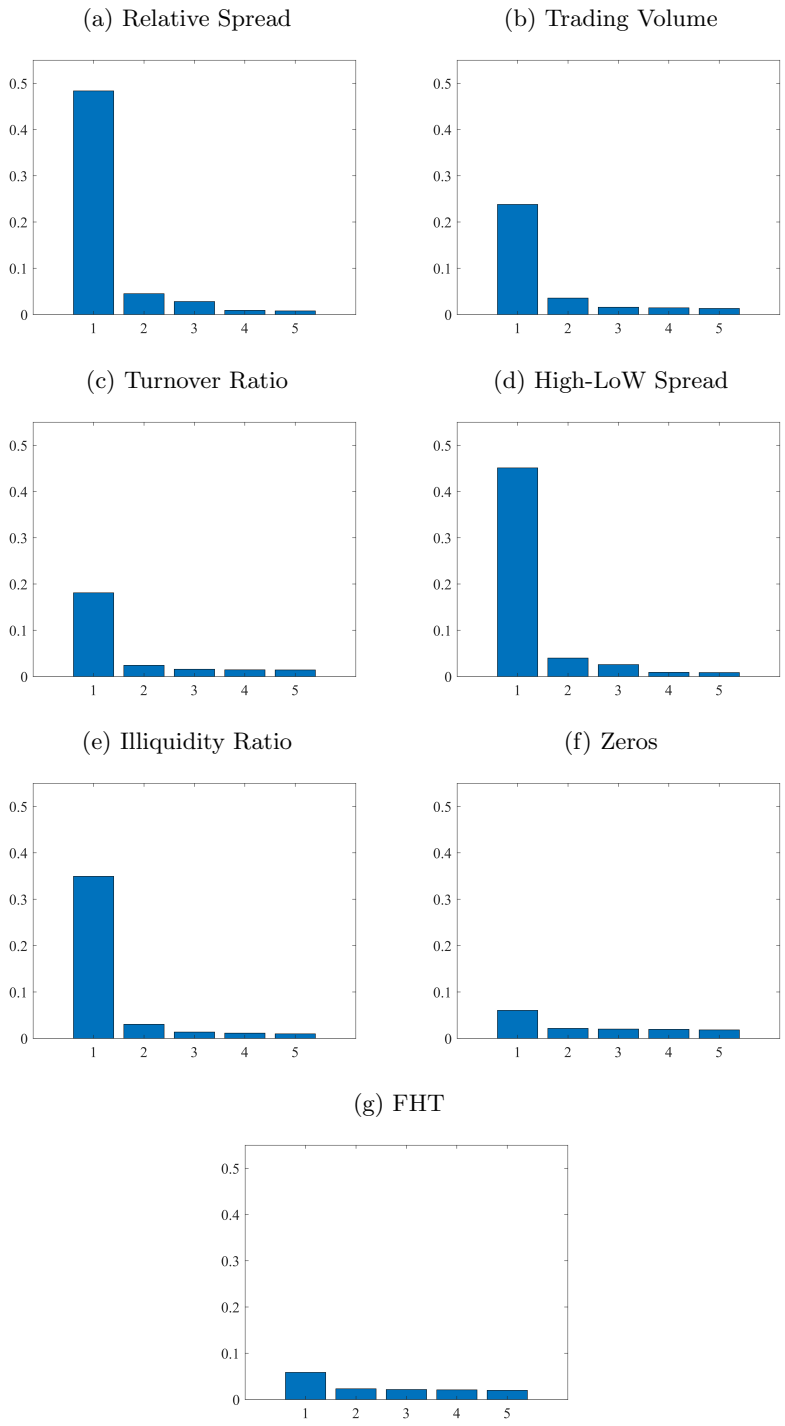
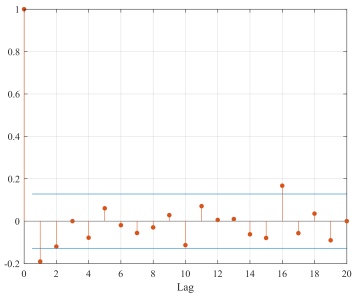


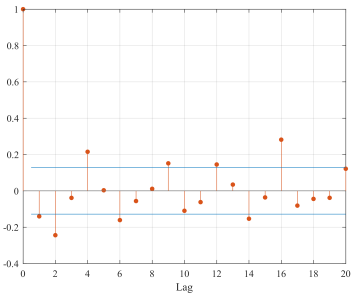
Figure 3: Fraction of the total variance explained by the first five estimated factors.

These figures show the fraction of the total variance explained by the first five estimated factors for each of our liquidity measure. Specifically, we report the eigenvalue ratio of each of the estimated factors.

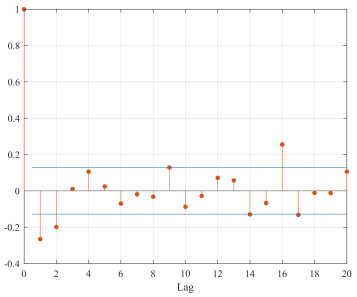
(a) Relative Spread



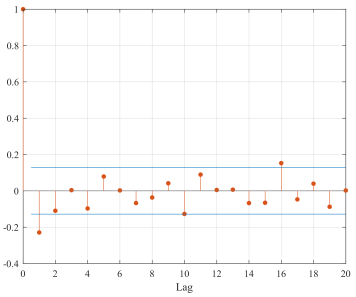
(b) Trading Volume



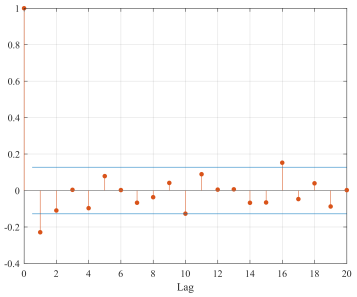
(c) Turnover Ratio



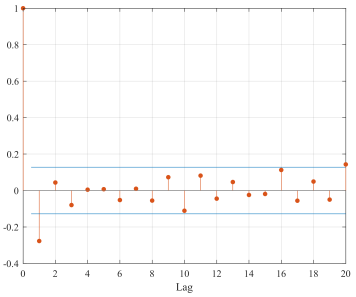
(d) High-Low Spread



(e) Illiquidity Ratio



(f) Zeros



(g) FHT

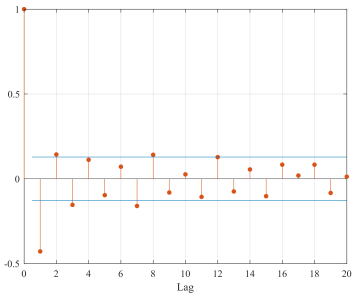


Figure 4: Autocorrelation of the estimated factors (first factor only).

These figures show the autocorrelation of the estimated first factors.

Table 4: In-Sample: Prediction of GDP Growth

	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Panel A: Forecast Horizon: h=1							
Coefficient (×1000)							
Factor 1	0.72	3.90**	-2.09	3.87**	4.54**	0.21	4.51
Factor 2	-8.81	3.36		-10.75	-0.5		
Factor 3	0.06			1.46			
R-squared	0.49%	9.27%	0.74%	9.8%	6.07%	0.06%	1.09%
AIC	-1.509	-1.529	-1.512	-1.531	-1.523	-1.510	-1.513
BIC	-1.499	-1.516	-1.505	-1.517	-1.513	-1.503	-1.506
Panel B: Forecast Horizon: h=2							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Factor 1	-0.53	3.40*	2.79	2.92*	4.64**	0.23	8.51
Factor 2	2.13	5.41		0.7	4.45		
Factor 3	5.06			2.2			
R-squared	2.26%	2.66%	0.71%	2.36%	3.97&	0.69%	2.49%
AIC	-1.333	-1.332	-1.331	-1.331	-1.337	-1.331	-1.336
BIC	-1.323	-1.318	-1.324	-1.317	-1.327	-1.324	-1.329
Panel C: Forecast Horizon: h=4							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Coefficient (×1000)							
Factor 1	3.56	6.51*	0.91	3.78	7.42**	1.77	12.58
Factor 2	7.94	6.22		5.08	6.71		
Factor 3	10.14			7.41			
R-squared	2.99%	3.11%	0.91%	2.53%	4.58%	0.91%	2.64%
AIC	-1.128	-1.126	-1.125	-1.125	-1.132	-1.125	-1.129
BIC	-1.118	-1.112	-1.118	-1.111	-1.122	-1.118	-1.122

This table includes the results of the prediction using individual stock market liquidity on next quarter's U.S. GDP growth controlling for current GDP growth. Panel A-C reports the coefficient associated with our liquidity factors. the R-squared, AIC, and BIC for different forecast horizons. A $\star$ and $\star\star$ indicates a significant coefficient at a 5-percent and 1-percent respectively.

Table 5: Granger-Causality Test: Liquidity Factors

	Relative Spread Relative Spread	Trading Volume Trading Volume	Turnover Ratio Turnover Ratio	High-Low High-Low	Illiquidity Ratio Illiquidity Ratio	Zeros Zeros	FTH FTH
Cause Liquidity	6.9	39.2**	2.11	36.78**	21.68**	0.01	13.38*
Cause GDP:							
Factor 1	2.95	3.31	1.05	4.42	1.03	0.05	0.83
Factor 2	0.46	7.95		0.66	1.95		
Factor 3	1.17			0.23			

This table includes the results of the Granger Causality between liquidity and GDP Growth. The columns contain the corresponding χ^2 statistics for different testing directions, whereas the first column provides the cause as well as the tested variable.. A $\star$ and $\star\star$ indicates a significant at a 5-percent and 1-percent respectively.

Table 6: Out-of-Sample: Liquidity vs. AR

Panel A: Forecast Horizon: h=1							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.03	0.97	1.03	0.95	0.94	1.02	1.02
MSE-F	-3.10	3.33**	-3.20	6.48**	7.62**	-2.17	-1.88
ENC-NEW	4.40	2.30*	0.23	6.40**	8.92**	-1.03	0.04
Panel B: Forecast Horizon: h=2							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.09	0.97	1.01	1.02	0.96	1.01	1.02
MSE-F	-10.19	2.28*	-1.19	-2.36	3.86**	-1.31	-2.25
ENC-NEW	-0.44	2.04**	-0.28	1.77	5.21**	-0.54	-0.14
Panel C: Forecast Horizon: h=4							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.14	0.97	1.05	1.08	0.97	1.00	1.01
MSE-F	-14.52	4.12**	-6.13	-8.86	4.14**	0.00	-0.73
ENC-NEW	-2.80	4.14**	-2.17	-2.53	4.16**	0.15	0.65

Panel A-C reports the out-of-sample results of comparing an AR model of US growth with liquidity to an AR-model. We present, MSE-Ratio, MSE-F test statistic, and the ENC-NEW test statistic. Entries below one for MSE-ratio indicate better performance than AR, while above one indicates that the AR performs better. MSFEs are calculated using rolling pseudo out-of-sample forecasts. The MSE-F statistic tests whether the unrestricted model (AR and stock market liquidity model) has better forecasting power than the restricted (AR model), whereas the ENC-NEW statistic tests if the restricted model encompasses the unrestricted model. A $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Table 7: Out-of-Sample: Comparing Predictive Accuracy

	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Panel A: Forecast Horizon: h=1							
Trading Volume	1.05*						
Turnover Ratio	1.00	0.95**					
High-Low	1.08*	1.03	1.08*				
Illiquidity Ratio	1.09**	1.03	1.09*	1.01			
Zeros	0.99	0.96**	1.01	0.93**	0.92**		
FTH	1.01	0.96**	1.01	0.93**	0.93**	1.00	2
Panel B: Forecast Horizon: h=2							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Trading Volume	1.13**						
Turnover Ratio	1.08*	0.96*					
High-Low	1.07*	0.95*	0.99				
Illiquidity Ratio	1.14**	1.01	1.05*	1.06			
Zeros	0.93**	0.96*	1.00	1.01	0.95*		
FTH	1.07*	0.95*	0.99	1.00	0.94*	0.99	
Panel C: Forecast Horizon: h=4							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
Trading Volume	1.18**						
Turnover Ratio	1.08*	0.92**					
High-Low	1.05	0.89**	0.98				
Illiquidity Ratio	1.18**	1.00	1.09*	1.11*			
Zeros	1.14*	0.97*	1.05	1.08*	0.97*		
FTH	1.13*	0.96*	1.05	1.07	0.96**	0.99	

Panel A-C reports the relative MSFEs between two models with different Liquidity measures. Entries below one for the ratio indicate better performance of model 1 (the column), whereas above one indicates a better performance of model 2 (the row). MSFEs are calculated using rolling pseudo out-of-sample forecasts. Using the modified Diebold-Mariano test, we check whether this ratio is significantly different from one, a $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Table 8: Out-of-Sample United Kingdom: Liquidity vs. AR

Panel A: Forecast Horizon: h=1							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	0.97	0.88	1.03	1.00	0.83	1.06	1.16
MSE-F	3.75**	15.82**	-2.83	-0.46	23.60**	-7.17	-16.85
ENC-NEW	2.65**	10.05**	0.18	0.19	14.36**	-3.48	-7.71
Panel B: Forecast Horizon: h=2							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.00	0.82	0.95	0.99	0.84	1.05	1.07
MSE-F	-0.15	25.88**	6.18**	0.47	22.03**	-5.80	-8.09
ENC-NEW	0.34	19.15**	3.42**	3.77*	15.82**	-2.55	-3.47
Panel C: Forecast Horizon: h=4							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.04	0.96	1.02	1.01	0.92	1.02	1.04
MSE-F	-4.52	5.49**	-2.32	-1.17	9.67**	-1.79	-4.03
ENC-NEW	-2.20	4.37**	-0.99	-0.12	8.79**	-0.85229	-1.5675

In this table, we present the Out-of-Sample results of forecasting real U.K. growth using U.S. liquidity. Panel A-C reports the out-of-sample results of comparing an AR model with liquidity to an AR-model. We present, MSE-Ratio, MSE-F test statistic, and the ENC-NEW test statistic. Entries below one for MSE-ratio indicate better performance than AR, while above one indicates that the AR performs better. MSFEs are calculated using rolling pseudo out-of-sample forecasts. The MSE-F statistic tests whether the unrestricted model (AR and stock market liquidity model) has better forecasting power than the restricted (AR model), whereas the ENC-NEW statistic tests if the restricted model encompasses the unrestricted model. A $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Table 9: Out-of-Sample Canada: Liquidity vs. AR

Panel A: Forecast Horizon: h=1							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.00	0.95	0.99	1.01	0.96	1.00	1.00
MSE-F	-0.38	2.23*	1.23	-0.75	2.13*	-0.50	-0.21
ENC-NEW	3.41*	2.03*	0.91	1.29	1.88*	-0.09	0.37
Panel B: Forecast Horizon: h=2							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.00	0.99	1.01	0.99	0.98	1.00	1.00
MSE-F	0.33	1.23	-1.21	0.68	2.35**	0.27	-0.42
ENC-NEW	0.41	2.87*	-0.10	1.55	2.98*	0.19	-0.02
Panel C: Forecast Horizon: h=4							
	Relative Spread	Trading Volume	Turnover Ratio	High-Low	Illiquidity Ratio	Zeros	FTH
MSE-Ratio	1.03	0.97	0.99	1.01	0.98	1.02	1.00
MSE-F	-2.91	3.30**	1.69	-1.42	2.34*	-1.98	0.23
ENC-NEW	-0.86	3.97**	1.48	1.49	2.62**	0.34	0.37

In this table, we present the Out-of-Sample results of forecasting real Canadian growth using U.S. liquidity. Panel A-C reports the out-of-sample results of comparing an AR model with liquidity to an AR-model. We present, MSE-Ratio, MSE-F test statistic, and the ENC-NEW test statistic. Entries below one for MSE-ratio indicate better performance than AR, while above one indicates that the AR performs better. MSFEs are calculated using rolling pseudo out-of-sample forecasts. The MSE-F statistic tests whether the unrestricted model (AR and stock market liquidity model) has better forecasting power than the restricted (AR model), whereas the ENC-NEW statistic tests if the restricted model encompasses the unrestricted model. A $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Table 10: Aggregate Liquidity: variance decomposition

	Common Factors	Individual Factors
Relative Spread	83.51	16.49
Trading Volume	89.86	10.14
Turnover Ratio	92.11	7.89
High-Low	83.21	16.79
Illiquidity Ratio	89.13	10.87
Zeros	63.46	36.54
FTH	64.80	35.20

This table includes the variance decomposition of our aggregate measure of liquidity. The first column states the average amount explained by the common factors for each liquidity measure and the second column the average amount of variance explained by orthogonalized individual factors.

Table 11: In-Sample: Prediction of GDP Growth with Aggregate Liquidity

Panel A: Forecast Horizon: h=1	
	Factor Model
Coefficient (×1000)	
Factor 1	6.48**
Factor 2	-4.55*
Factor 3	-1.21
R-squared	8.08%
AIC (÷1000)	-1.52665
BIC (÷1000)	-1.512
Panel B: Forecast Horizon: h=2	
	Factor Model
Coefficient (×1000)	
Factor 1	4.67**
Factor 2	-6.85*
Factor 3	4.13
R-squared	4.07%
AIC (÷1000)	-1.336
BIC (÷1000)	-1.322
Panel C: Forecast Horizon: h=4	
	Factor Model
Coefficient (×1000)	
Factor 1	5.96**
Factor 2	-12.10*
Factor 3	-4.83
R-squared	4.49%
AIC (÷1000)	-1.1130
BIC (÷1000)	-1.116

This table includes the results of the prediction using aggregate stock market (il)liquidity on next quarter's GDP growth controlling for current GDP growth. Panel A-C reports the coefficient associated with our (il)liquidity series, the R-squared, AIC, and BIC for different forecast horizons. A $\star$ and $\star\star$ indicates a significant coefficient at a 5-percent and 1-percent respectively.

Table 12: Granger-Causality Test: Aggregate Liquidity

Panel A: Forecast Horizon: h=1	
	Factor Model
Cause Liquidity	43.53**
Cause GDP:	
Factor 1	5.44
Factor 2	6.05
Factor 3	2.46

This table includes the results of the Granger Causality between liquidity and GDP Growth. The columns contain the corresponding χ^2 statistics for different testing directions. A $\star$ and $\star\star$ indicates a significant at a 5-percent and 1-percent respectively.

Table 13: Out-of-Sample: Aggregate Liquidity vs. AR

Panel A: Forecast Horizon: h=1	
	Factor Model
MSE-Ratio	0.93
MSE-F	8.88**
ENC-NEW	8.77**
Panel B: Forecast Horizon: h=2	
MSE-Ratio	0.96
MSE-F	4.40**
ENC-NEW	5.46**
Panel C: Forecast Horizon: h=4	
MSE-Ratio	0.95
MSE-F	9.52**
ENC-NEW	11.47**

This table includes the results of the out-of-sample forecasting of real U.S. GDP growths using measures of aggregate liquidity- Panel A-C reports the relative MSFEs with respect to AR, MSE-F test statistic and the ENC-NEW test statistic for different forecast horizons. Entries below one for MSE-ratio indicate better performance than AR. MSFEs are calculated using rolling pseudo out-of-sample forecasts. The MSE-F statistic tests whether the unrestricted model (AR and stock market liquidity model) has better forecasting power than the restricted (AR model), whereas the ENC-NEW statistic tests if the restricted model encompasses the unrestricted model. A $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Table 14: Out-of-Sample: Comparing Predictive Accuracy Aggregate vs. Individual Liquidity

Panel A: Forecast Horizon: h=1	
	Factor Model
Relative Spread	1.10**
Trading Volume	1.05
Turnover Ratio	1.10**
High-Low	1.02
Illiquidity Ratio	0.98*
Zeros	1.09*
FTH	1.09*
Panel B: Forecast Horizon: h=2	
	Factor Model
Relative Spread	1.14**
Trading Volume	1.01
Turnover Ratio	1.05*
High-Low	1.06*
Illiquidity Ratio	1.00
Zeros	1.05*
FTH	1.06*
Panel C: Forecast Horizon: h=4	
	Factor Model
Relative Spread	1.19**
Trading Volume	1.00
Turnover Ratio	1.10*
High-Low	1.13*
Illiquidity Ratio	1.00
Zeros	1.04*
FTH	1.06*

Panel A-C reports the relative MSFEs between aggregate liquidity and different Liquidity measures and in brackets the Modified Diebold-Mariano statistic for different forecast horizons. Entries below one for the ratio indicate better performance of model 1 (the row), whereas above one indicates a better performance of model 2 (the column). MSFEs are calculated using rolling pseudo out-of-sample forecasts. A $\star$ and $\star\star$ indicate a significant test statistic at a 5-percent level and 1-percent respectively.

Appendix A: Construction of Liquidity measures

Relative Spread

The relative spread is constructed as the absolute difference between bid-ask prices divided by the midpoint of the bid-ask prices:

$$\frac{p^a - p^b}{\frac{p^a + p^b}{2}}, \tag{10}$$

where p^a is the ask-price and p^b the bid-price. By dividing the bid-ask spread by the midpoint of the bid-ask prices, we can easily compare the bid-ask spread and eliminate the effect of the difference in the size of bid-ask prices.

Trading Volume

The trading volume provides the dollar amount of shares traded in a quarter. It is calculated on a daily basis by multiplying the price of a stock by the volume traded:

$$TV = V \times P, \tag{11}$$

where V is volume and P is the price.

Turnover Ratio

The turnover ratio is calculated as the sum of volume traded over the quarter divided by the amount of shares outstanding on the last trading day of the quarter:

$$TR = \frac{V^q}{SO}, \tag{12}$$

where V^q is the volume traded over a quarter and SO the share outstanding.

High-Low Spread

The high-low estimator is constructed as follows:

$$High - Low_{i,t} = \frac{2(e^{\alpha_{i,t}} - 1)}{(1 + e^{\alpha_{i,t}})} \tag{13}$$

where $\alpha_{i,t}$ is defined as

$$\alpha_{i,t} = \frac{2\sqrt{\beta_{i,t}} - \sqrt{\beta_{i,t}}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma_{i,t}}{3 - 2\sqrt{2}}} \tag{14}$$

$$\tag{15}$$

with

$$\beta_{i,t} = \mathbf{E}\left[\sum_{j=0}^{1}\left(ln\left(\frac{H_{t+j,i}}{L_{t+j,i}}\right)\right)^2\right] \tag{16}$$

$$\gamma_{i,t} = \left[ln\left(\frac{H_{t,t+1,i}}{L_{t,t+1}}\right)^2\right] \tag{17}$$

where $H_{t,i}$ and $L_{t,i}$ refer to high- and low-price at day t of firm i and $H_{t,t+1,i}$ and $L_{t,t+1,i}$ to high and low-prices over days t and $t+1$.

Illiquidity Ratio

The illiquidity ratio is constructed as the absolute return on a day divided by the dollar amount traded on the same day:

$$ILR = \frac{|r|}{V \times P} \tag{18}$$

where $|r|$ is the absolute return, V volume traded, and P the stock price. A high ratio means that a stock is illiquid, stating that a stock is illiquid when its prices fluctuate while there is little trading.

Zeros

The zeros liquidity uses the fraction of zero return trading days over a trading period:

$$Z = \frac{\text{Number of days with zero return}}{\text{Number of trading days in a month}}. \tag{19}$$

FHT

FHT is calculated as follows:

$$2\sigma N^{-}1\left(\frac{1+z}{2}\right), \tag{20}$$

where σ is a stock standard deviation over the quarter, N^{-1} the inverse of the cumulative normal distribution, and z the ratio of zero return days over total days in a quarter.